Mukul Kumar

Software Engineer | Backend Development & AI Applications

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Education

Indian Institute of Technology Hyderabad — Aug 2022 – May 2026
Bachelor of Technology in Chemical Engineering — *GPA: 6.85 / 10*

Work Experience

People Tech Group Inc — Jun – Aug 2025
Software Developer Intern – Software-Defined Vehicles — *Hyderabad, India*

- Built a **C++ middleware bridge** to transfer vehicle-signal data from **SOME/IP-based ECUs** to the **KUKSA Databroker** for software-defined vehicle applications.
- Implemented a **VSOMEIP adapter** to parse binary vehicle-signal payloads and convert them into typed **Protocol Buffers messages** for downstream communication.
- Integrated and tested vehicle-signal communication across multiple SDV modules using **VSOMEIP and KUKSA**, with **CMake** for build configuration.

Open Source Contributions

Budibase – Open Source Contributor — Sep 2026
TypeScript, Node.js, Jest — Merged PR #19667

- Fixed a bug where applications continued requesting deleted or unused plugins, causing **404 errors** during application loading.
- Updated plugin tracking to **remove unused references when publishing** and filter out uninstalled plugins before loading, while preserving plugins still used by other screens.
- Added **automated regression tests** covering component removal, screen deletion, and shared plugin usage; contribution merged through **PR #19667**.

Projects

DocuMind AI — Jul – Aug 2025
Python, FastAPI, Gemini, OpenAI Embeddings, Azure TTS, Docker — GitHub

- Built a document intelligence platform supporting **PDF ingestion, semantic search, and audio-based Podcast Mode** using **FastAPI**.
- Integrated **OpenAI embeddings** for semantic retrieval, **Google Gemini** for content generation, and **Azure TTS** for audio generation.
- Ranked among the **Top 100 teams** in the **Adobe India Hackathon 2025**, which attracted **60,000+ participants**.

Technical Skills

- **Languages:** C++, Java, Python, TypeScript, SQL
- **Backend:** Spring Boot, FastAPI, Node.js, REST APIs
- **Databases:** MySQL, PostgreSQL
- **Communication & Protocols:** gRPC, Protocol Buffers, SOME/IP
- **Developer Tools & Testing:** Docker, Linux, Git, GitHub, Postman, CMake, Jest
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems

Achievements

- **Problem Solving:** Solved **450+ problems** across LeetCode and Codeforces; achieved a **peak LeetCode rating of 1671**.
- **National Talent Search Examination (NTSE):** Qualified Stage I and Stage II.
- **Sports:** Represented IIT Hyderabad in volleyball at the Inter IIT Sports Meet.